

GitHub

PYTHON COURSE 2026

What is Git(Hub)?

Git ([/git/](https://git/))^[8] is a [distributed version control system](#)^[9] that tracks versions of [files](#). It is often used to control [source code](#) by [programmers](#) who are [developing](#) software collaboratively.

(Wikipedia)

- Open source
- Originally created by Linus Torvald for Linux development
- Known through platforms like GitHub (Microsoft) and GitLab (GitLab Inc.)
- “Repository” as key structure
- Many different ways to access a repository
(command line, GUI clients, website, IDEs ...)



GitHub web interface

  eulerlab / QDSpy

[Code](#) [Issues 6](#) [Pull requests](#) [Discussions](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

 **QDSpy** Public

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[experimental](#) [3 Branches](#) [4 Tags](#)

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This branch is **13 commits ahead of** `master`.

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 **teuler** Added a new option to play videos (internally) 7544524 · yesterday 526 Commits

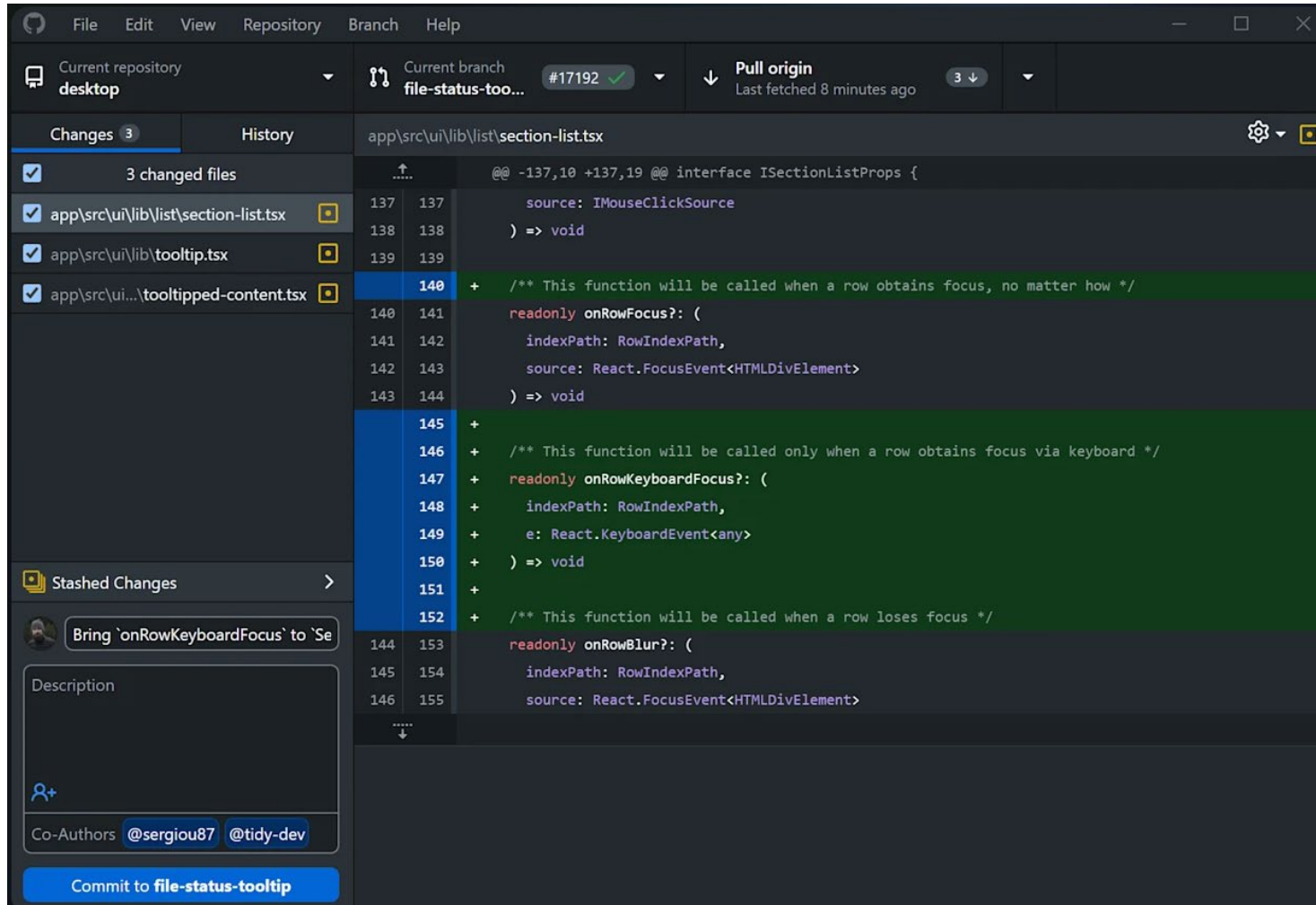
Arduino	Create QDSpy_DIO_wTrigger.ino	3 years ago
Devices	Fixing bugs related to video stimuli	yesterday
Graphics	Some bug fixes in rendering/GUI (RPi)	last month
Libraries	Fix MQTT	3 months ago
Release_notes	MQTT version (experimental)	10 months ago
Shader	Distort shader (experimental)	2 months ago
Sounds	New features (progress, sound) and bug fixes	last year
Stimuli	Fixing bugs related to video stimuli	yesterday
.gitignore	Squashed commit of the following:	last month
.python-version	Now using uv for package management	last month
Disable_ExtensiveGammaChanges.reg	Version 0.6b	9 years ago

[qdspsy.eulerlab.de/](#)
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Releases 4
 **QDSpy v.95** Latest
on May 25
[+ 3 releases](#)

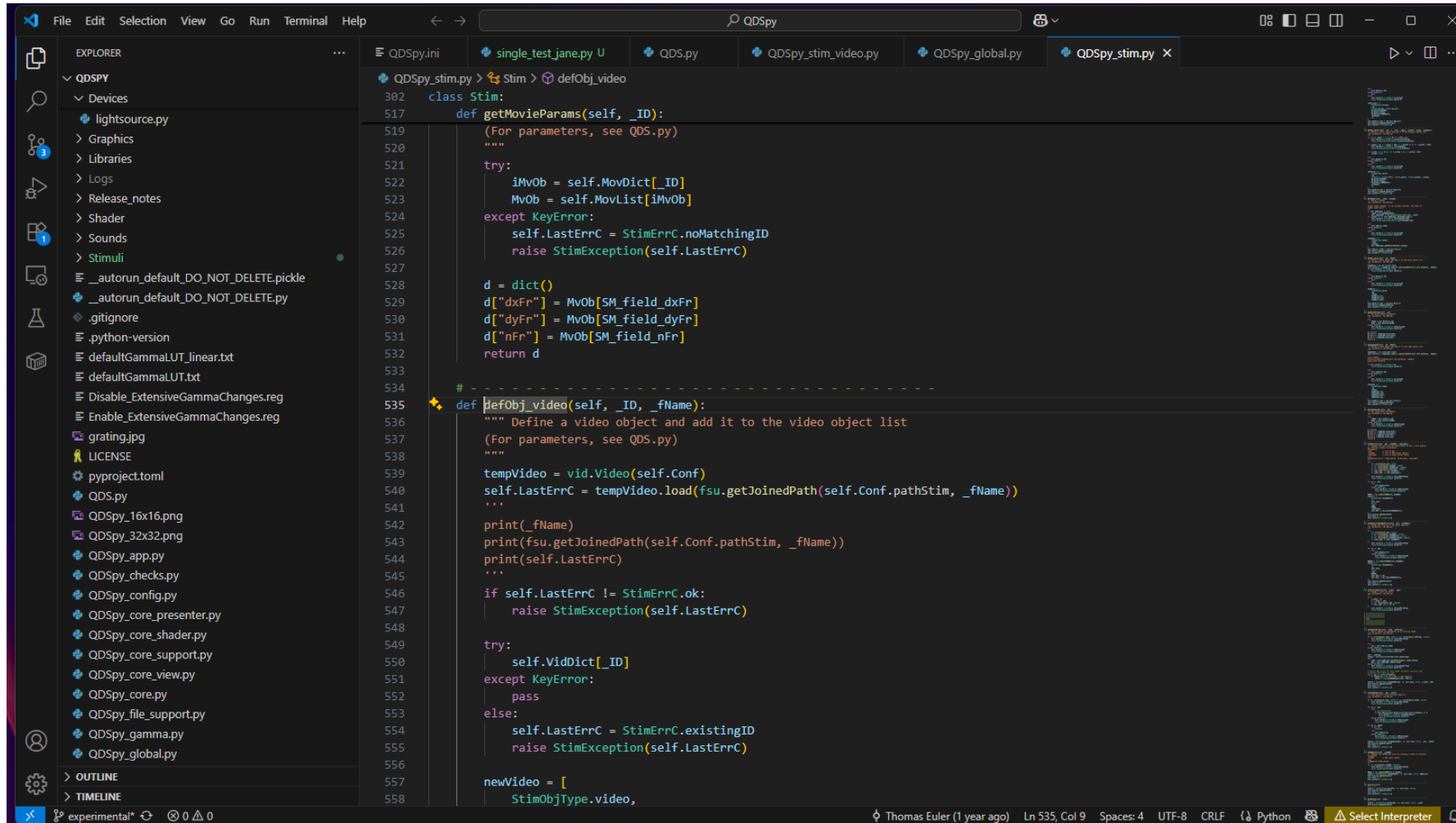
GitHub Desktop



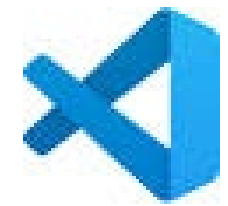
Repository management
Mac and Windows



Visual Studio Code (VSCode)



IDE
all platforms



Creating a new GitHub repository

- create a new repository in your GitHub account on the website
- no template
- name it and give it a brief description
- you can make it „public“ or „private“ (can be changed to „public“ later)
- check add README file
- choose a „.gitignore template“
- choose a license
- click „create“ and you are done

Adding information to your new repository

- edit the README file (markdown format, like in Jupyter notebooks)
- press „commit“ – for now just commit directly to the main branch
- add a new text file (.md for markdown) ...
- create a link to the new file in README
(link format: „[instructions] (https://github.com/teuler/mytest/blob/main/instructions) “)

Cloning the repository and adding file(s)

- you can clone the repository to your local computer, but for testing, you can also use your bwJupyter account, as described here:
- Start a terminal in the byJupyter „Launcher“ tab
- change into your work directory
`cd work`
- clone your repository:
`git clone https://github.com/teuler/mytest.git`
- Change into directory
`cd mytest`
- Now your „local“ repository is in the work folder.
Add a short Python program and test-run it in the terminal.

Committing/pushing changes to the repository

- to do once:
 - make sure that git knows you:

```
git config --global user.email "thomas@eulerlab.de"
git config --global user.name "teuler"
```
 - create a token – it replaces the password
(see <https://github.com/settings/tokens>; create a “classic” token and store it – you cannot retrieve it!)
- add last changes to the commit (“stage” is synonymous with “add”)

```
git add *
```
- commit changes to your local repository; a message is mandatory!

```
git commit -m "Python test file added"
```
- push the changes to the repository on GitHub

```
git push
```

(note that you need to enter your GitHub user name and token [instead of a password]).

Pulling changes to the local repository

- if the repository is public, you just need:

```
git pull
```

(note that for private repositories, you need access and then enter your GitHub user name and token [instead of the password]).

- if there have been changes both locally and remotely, it becomes a bit more complicated. you have to decide how to solve potential conflicts (keywords are here “stash” and “merge”).
- If you don't use git commands that often, tools like GitHub Desktop facilitate handling issues a lot

Pull requests, issues, branches ...

- pull request allow repository teams to review changes before including them (can all be managed in the web interface or with tools like Github Desktop or VSCode ...)
- issues represent a mechanism to alert the repository team of potential problems, including documentation and history
- branches allow several versions of the repository in parallel (and enable a later integration)
- releases “freeze” the current state of a repository (e.g., a version of a software)

Authentication

- personal tokens
- pass key via the GitHub website
- Two-factor authentication via the GitHub app

INSTALLATION & GUIs

With platform specific installers for Git, GitHub also provides the ease of staying up-to-date with the latest releases of the command line tool while providing a graphical user interface for day-to-day interaction, review, and repository synchronization.

GitHub for Windows
https://windows.github.com

GitHub for Mac
https://mac.github.com

For Linux and Solaris platforms, the latest release is available on the official Git web site.

Git for All Platforms
http://git-scm.com

SETUP

Configuring user information used across all local repositories

git config --global user.name "[firstname lastname]"
set a name that is identifiable for credit when review version history
git config --global user.email "[valid-email]"
set an email address that will be associated with each history marker
git config --global color.ui auto
set automatic command line coloring for Git for easy reviewing

SETUP & INIT

Configuring user information, initializing and cloning repositories

git init
initialize an existing directory as a Git repository
git clone [url]
retrieve an entire repository from a hosted location via URL

STAGE & SNAPSHOT

Working with snapshots and the Git staging area

git status
show modified files in working directory, staged for your next commit
git add [file]
add a file as it looks now to your next commit (stage)
git reset [file]
unstage a file while retaining the changes in working directory
git diff
diff of what is changed but not staged
git diff --staged
diff of what is staged but not yet committed
git commit -m "[descriptive message]"
commit your staged content as a new commit snapshot

BRANCH & MERGE

Isolating work in branches, changing context, and integrating changes

git branch
list your branches. a * will appear next to the currently active branch
git branch [branch-name]
create a new branch at the current commit
git checkout
switch to another branch and check it out into your working directory
git merge [branch]
merge the specified branch's history into the current one
git log
show all commits in the current branch's history

INSPECT & COMPARE

Examining logs, diffs and object information

git log
show the commit history for the currently active branch
git log branchB..branchA
show the commits on branchA that are not on branchB
git log --follow [file]
show the commits that changed file, even across renames
git diff branchB...branchA
show the diff of what is in branchA that is not in branchB
git show [SHA]
show any object in Git in human-readable format

TRACKING PATH CHANGES

Versioning file removes and path changes

git rm [file]
delete the file from project and stage the removal for commit
git mv [existing-path] [new-path]
change an existing file path and stage the move
git log --stat -M
show all commit logs with indication of any paths that moved

IGNORING PATTERNS

Preventing unintentional staging or committing of files

logs/ *.notes pattern*/
Save a file with desired patterns as .gitignore with either direct string matches or wildcard globs.
git config --global core.excludesfile [file]
system wide ignore pattern for all local repositories

SHARE & UPDATE

Retrieving updates from another repository and updating local repos

git remote add [alias] [url]
add a git URL as an alias
git fetch [alias]
fetch down all the branches from that Git remote
git merge [alias]/[branch]
merge a remote branch into your current branch to bring it up to date
git push [alias] [branch]
Transmit local branch commits to the remote repository branch
git pull
fetch and merge any commits from the tracking remote branch

REWRITE HISTORY

Rewriting branches, updating commits and clearing history

git rebase [branch]
apply any commits of current branch ahead of specified one
git reset --hard [commit]
clear staging area, rewrite working tree from specified commit

TEMPORARY COMMITS

Temporarily store modified, tracked files in order to change branches

git stash
Save modified and staged changes
git stash list
list stack-order of stashed file changes
git stash pop
write working from top of stash stack
git stash drop
discard the changes from top of stash stack

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Learn how to start building, shipping, and maintaining software with GitHub. Explore our products, sign up for an account, and connect with the world's largest development community.

Quickstart

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[About GitHub and Git](#)

You can use GitHub and Git to collaborate on work.

[GitHub's plans](#)

An overview of GitHub's pricing plans.

[Getting started with your GitHub account](#)

With a personal account on GitHub, you can import or create repositories, collaborate with others, and connect with the GitHub community.

[Getting started with GitHub Team](#)

With GitHub Team groups of people can collaborate across many projects at the same time in an organization account.

Popular

[Creating an account on GitHub](#)

Create a personal account to get started with GitHub.

[Hello World](#)

Follow this Hello World exercise to learn GitHub's pull request workflow.

[Set up Git](#)

At the heart of GitHub is an open-source version control system (VCS) called Git. Git is responsible for everything GitHub-related that happens locally on your computer.

[About versions of GitHub Docs](#)

You can read documentation that reflects the GitHub product you're currently using.